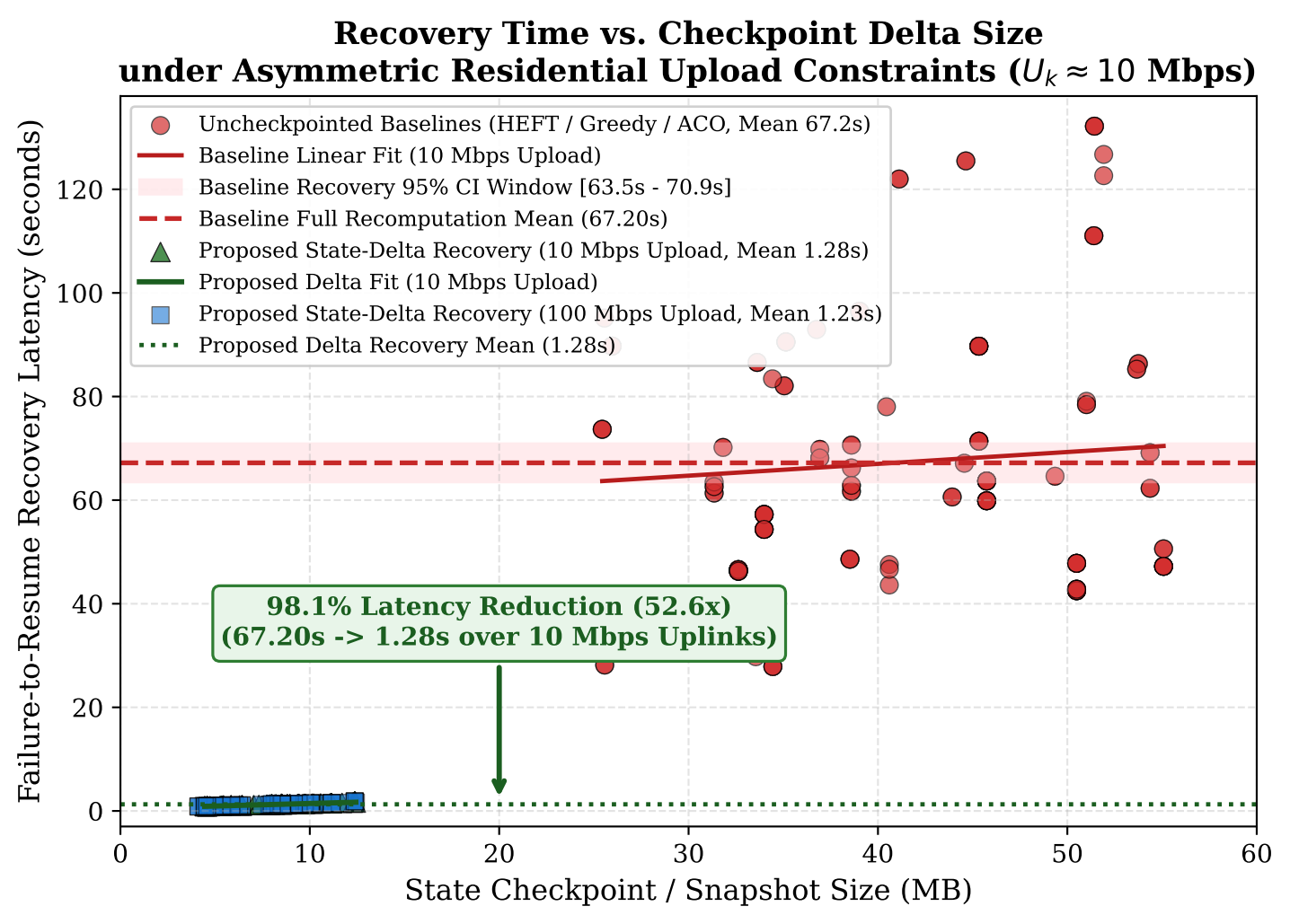
Recovery Time vs. Checkpoint Delta Size
under Asymmetric Residential Upload Constraints ($U_k \approx 10$ Mbps)
Uncheckpointed Baselines (HEFT / Greedy / ACO, Mean 67.2s)
Baseline Linear Fit (10 Mbps Upload)
Baseline Recovery 95% CI Window [63.5s - 70.9s]
Baseline Full Recomputation Mean (67.20s)
Proposed State-Delta Recovery (10 Mbps Upload, Mean 1.28s)
Proposed Delta Fit (10 Mbps Upload)
Proposed State-Delta Recovery (100 Mbps Upload, Mean 1.23s)
Proposed Delta Recovery Mean (1.28s)
98.1% Latency Reduction (52.6x)
(67.20s -> 1.28s over 10 Mbps Uplinks)
Failure-to-Resume Recovery Latency (seconds)
State Checkpoint / Snapshot Size (MB)
0
20
40
60
80
100
120
0
10
20
30
40
50
60